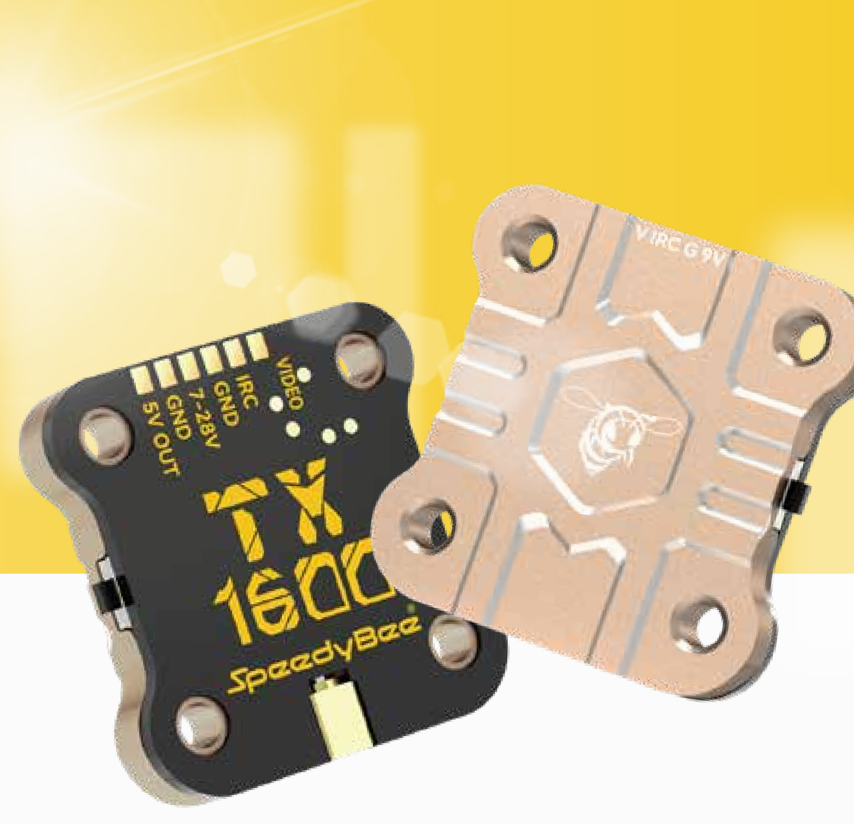
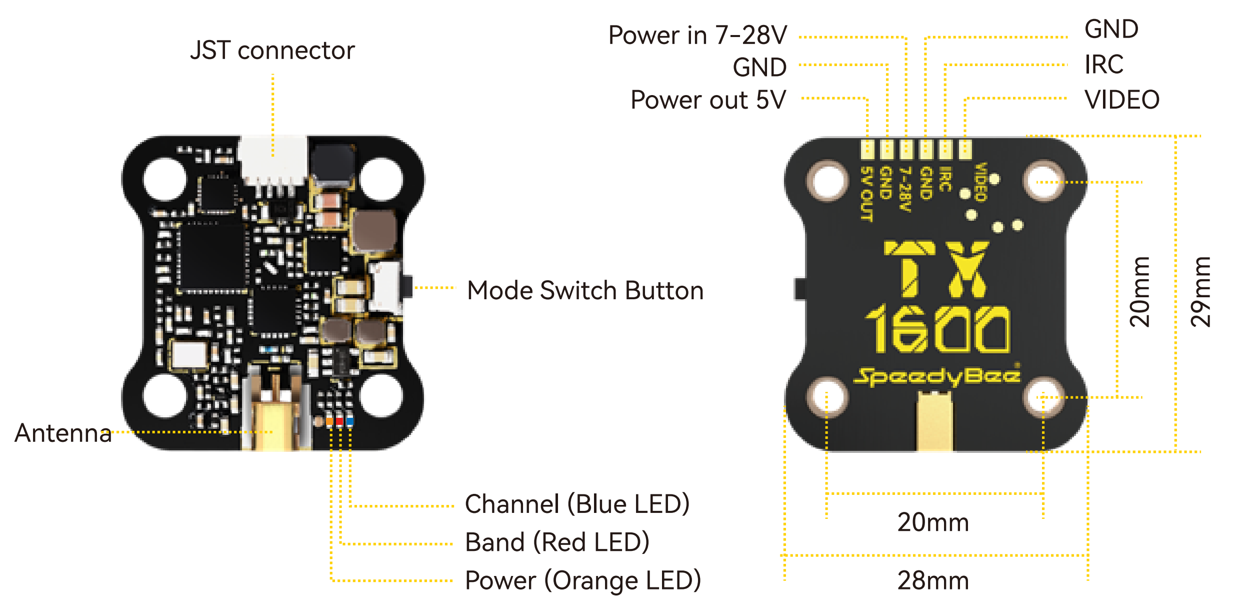
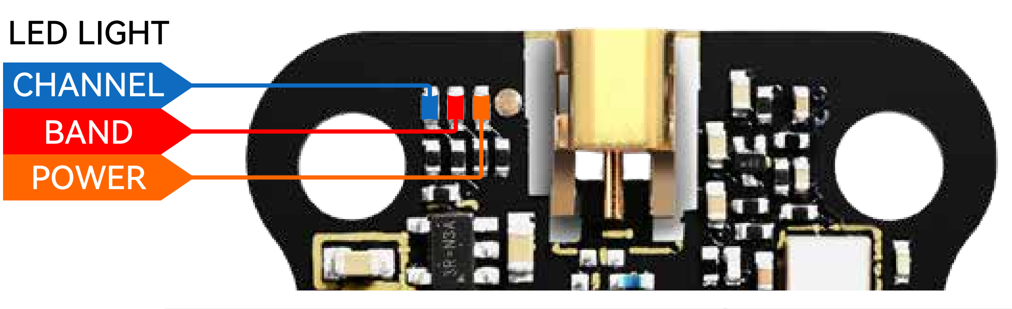


TX ULTRA USER MANUAL

Instruction Diagram



Notice: Please use the screws to install and fix the TX ULTRA as the mainboard and the cover are separated.



	Lock	Unlock
Blue LED Channel	**Constantly ON:** The current frequency point is Channel 1 **OFF:** The current frequency point is one of Channel 2~Channel 8.	
Red LED Band	Red LED **will keep blinking**, at this time the Red LED is not used to indicate the current Band information.	**Constantly ON:** the current band is Band1, **OFF:** the current band is one of Band 2 ~ Band 6.
Orange LED Power	Orange LED **will keep blinking**, at this timethe Orange LED is not used for indicating the current Power information.	Light OFF — 25mW Blinking once in 3 seconds — 200mW Blinking once in 2 seconds — 800mW Blinking once in 1 second — 1000 mW Solid ON — Max mW Blinking 4 times per second (fast flash) — PIT mode

Key Operation

Short press the button to switch CH1/2/3/4/5/6/7/8.

Long press the button for 2 seconds to enter the BAND mode; then short press to switch band1/2/3/4/5/6.

Long press the button for 4 seconds to enter the POWER mode; then short press to switch to 25/200/800/Max mW.

Long press the button for 10 seconds to unlock/lock.

Support IRC Tramp Protocol

TX ULTRA supports the IRC TRAMP protocol and the modification of video transmission parameters through the remote control, including frequency, working power and so on. If your flight controller supports Bluetooth or WIFI, you can also modify the video transmission parameters through the SpeedyBee app.

NOTE: For BetaFlight flight controller firmware above BetaFlight 4.1.0,you need upload a VTX Table to the flight controller before using the remote control to modify the video transmitter parameters normally.

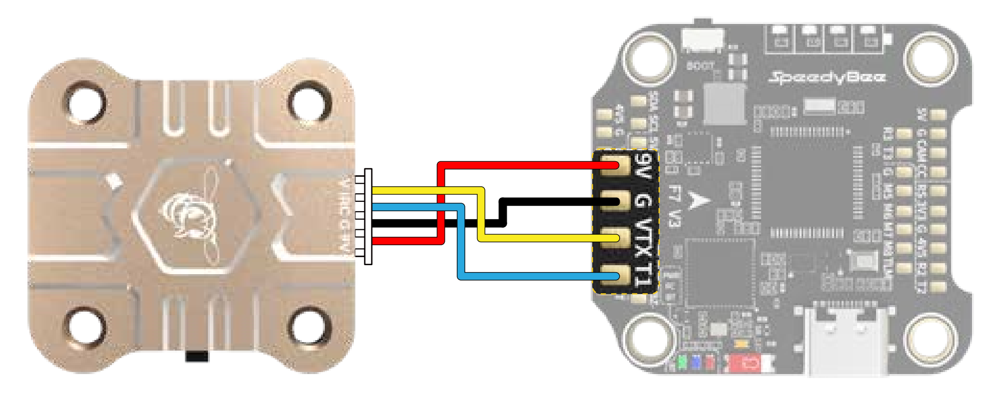
SpeedyBee-TX ULTRA（CE）.json"
SpeedyBee-TX ULTRA（USA）.json

For the use of the video transmitter table, please refer to this article "How to Setup Betaflight VTX Table – SmartAudio Tramp VTX Control" by Oscar Liang

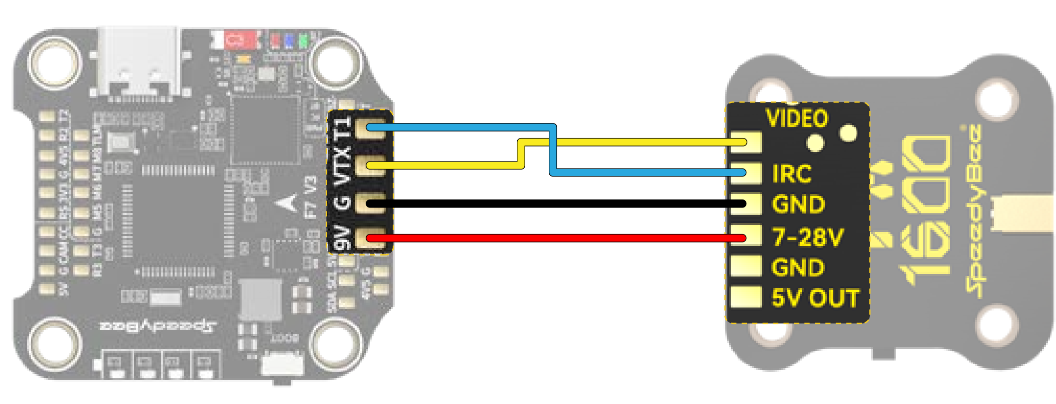
1. Flight controller wiring

Note: Take UART1 for example

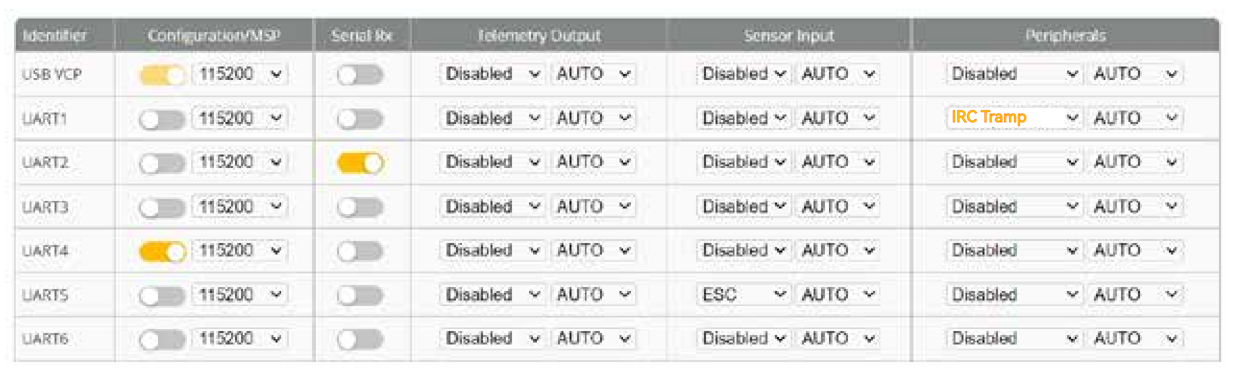
JST Connection:



Pad Connection:



2.BetaFlight 4.1 or above setting

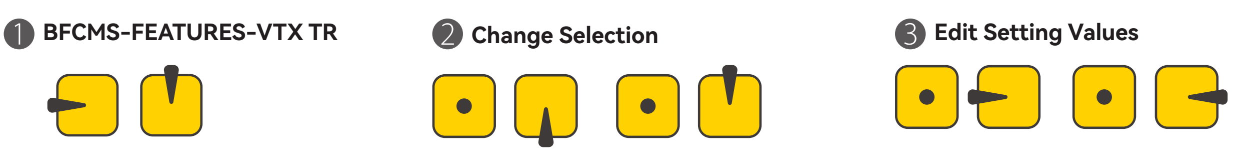


Identifier	Configuration/MSP		Serial Rx	Telemetry Output		Sensor Input		Peripherals	
USB VCP	on	115200	off	Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART1	off	115200	off	Disabled	AUTO	Disabled	AUTO	IRC Tramp	AUTO
UART2	off	115200	on	Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART3	off	115200	off	Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART4	on	115200	off	Disabled	AUTO	Disabled	AUTO	Disabled	AUTO
UART5	off	115200	off	Disabled	AUTO	ESC	AUTO	Disabled	AUTO
UART6	off	115200	off	Disabled	AUTO	Disabled	AUTO	Disabled	AUTO

3. Remote control

Note: Take Mode-2 for example

1 BFCMS-FEATURES-VTX TR　　2 Change Selection　　3 Edit Setting Values



Frequency Table

Orange is a disabled channel in locked state

Channel	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
1 Band A	5865	5845	5825	5805	5785	5765	5745	5725
2 Band B	5733	5752	5771	5790	5809	5828	5847	5866
3 Band E	5705	5685	5665	5645	5885	5905	5925	5945
4 Airwave	5740	5760	5780	5800	5820	5840	5860	5880
5 Race Band	5658	5695	5732	5769	5806	5843	5880	5917
6 Low Race	5362	5399	5436	5473	5510	5547	5584	5621